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Illusions of truth—Experimental insights into human and algorithmic detections of fake online reviews

Daria Plotkina[a,*], Andreas Munzel[b], Jessie Pallud[a]

[a] University of Strasbourg, EM Strasbourg Business School, HuManiS (EA 7308), 61 Avenue de la Forêt Noire, 67000 Strasbourg, France
[b] Toulouse University, UT1, Toulouse School of Management, TSM Research (CNRS), 2 rue du Doyen Gabriel Marty, 31042 Toulouse Cedex 9, France

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ABSTRACT

The issue of fake online reviews is increasingly relevant due to the growing importance of online reviews to consumers and the growing frequency of deceptive corporate practices. It is, therefore, necessary to be able to detect fake online reviews. An experiment with 1041 respondents allowed us to create two pools of reviews (fake and truthful) and compare them for psycholinguistic deception cues. The resulting automated tool accounted for review valence and incentive and detected deceptive reviews with 81% accuracy. A follow-up experiment with 407 consumers showed that humans have only a 57% accuracy of detection, even when a deception mindset is activated with information on cues of fake online reviews. Therefore, micro-linguistic automated detection can be used to filter the content of reviewing websites to protect online users. Our independent analysis of reviewing websites confirms the presence of dubious content and, therefore, the need to introduce more sophisticated filtering approaches.

1. Introduction

> "*He who tries to protect himself from deception is often cheated, even when most on his guard.*"
>
> *Plautus*

Deception, or deceptive communication, understood as "the deliberate attempt, whether successful or not, to conceal, fabricate, and/or manipulate in any other way factual and/or emotional information, by verbal and/or nonverbal means, in order to create or maintain in another or in others a belief that the communicator himself or herself considers false" (Masip, Garrido, & Herrero, 2004, p. 148), is a complex topic that is ever-present in human society and, thus, always relevant. One form of deception in the online environment, deceptive online reviews, has recently attracted increasing attention from academia, management, and the broader public (Deahl, 2018; Malbon, 2013).

The development of Internet communication platforms has made positive and negative product- and company-related statements made by potential, actual, and former customers an increasingly important source of information for consumers (Hennig-Thurau, Gwinner, Walsh, & Gremler, 2004, p. 39). Previous research shows that online reviews affect sales (Fan, Che, & Chen, 2017; Liu, Huang, & Zhang, 2016; Zhu & Zhang, 2010), purchase intentions (Park, Lee, & Han, 2007), and product choices (Fagerstrøm, Ghinea, & Sydnes, 2016). Because of their growing influence, online reviews are becoming increasingly relevant to marketers (Chen & Xie, 2008; Chong, Li, Ngai, Ch'ng, & Lee, 2016). The empirically supported effects of reviews on consumers and the difficulty in obtaining reviews have led companies to engage in malicious practices to manage their online reputations (Munzel, 2016). These corporate practices to manipulate opinions by posting deceptive content are referred to as deceptive online communication, or "fake reviews" (Mayzlin, Dover, & Chevalier, 2014). Although fake online reviews are illegal in many countries, they still exist, with companies ready to take risks to deceive their customers. For instance, in 2017, nine California car dealerships paid $3.6 million to settle charges brought by the Federal Trade Commission (FTC) for posting fake online reviews (Davis, 2017).

Deception poses a problem for both society and the marketplace. It undermines trust, and the consequences are costly (e.g., Keep & Schneider, 2010). When consumers have doubts about the trustworthiness of a review, a positive review can have a negative impact on purchase intentions (Reimer & Benkenstein, 2016). As a potential solution, deception detection makes it possible to evaluate reviews and coordinate further action against deceivers (e.g., confront or stop them). To detect deceit, however, we must first understand how deception is constructed and communicated.

Truthful information and deceptive information are dissimilar. On a subconscious level, different parts of the human brain are used for

* Corresponding author.
E-mail addresses: daria.plotkina@em-strasbourg.eu (D. Plotkina), andreas.munzel@tsm-education.fr (A. Munzel), jessie.pallud@em-strasbourg.eu (J. Pallud).

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being truthful and for lying, enabling lie detection via functional magnetic resonance imaging (fMRI) (Davatzikos et al., 2005). These differences cause distinct behavioral and linguistic patterns for deceptive and truthful communication. The linguistic characteristics that define communication and help in understanding its nature are called signs or cues. Psycholinguistics, which deals with the factors of language use and production, has been shown to be a potential approach for detecting deception (Depaulo et al., 2003; Ott, Choi, Cardie, & Hancock, 2011). At the same time, prior work has demonstrated that humans have a very limited capacity for detecting deception because of a significant truth bias (Levine, Park, & McCornack, 1999). Thus, the present research aims to expand on previous studies by finding psycholinguistic deception cues with a detection accuracy exceeding those of existing methods. We seek to create easily applicable guidelines to spot fake reviews and to test whether humans can detect deceit or whether they need support to do so. Our main research question is the following: Do humans need algorithmic tools to detect fake online reviews with significant accuracy?

To answer this research question, we conducted three studies. The focus of our first study was to learn how deceptive reviews are composed by examining two questions: (1) Is there any difference between truthful and fake reviews? (2) Can this difference be used to detect the fake messages? The second study analyzed consumers' capability to detect fake online reviews, while the third field study examined the occurrence of potentially fake online reviews on websites with restaurant reviews.

The present study contributes to the literature from multiple viewpoints. First, we add to the knowledge on the psycholinguistic deception cues of online reviews. We do so by applying a sophisticated linguistic analysis tool that has not previously been used in the online deception detection literature (i.e., Coh-Metrix) and by creating verified samples of both truthful and fake reviews. Second, we show that an automated algorithm can detect fake online reviews regardless of the review valence and reviewer motivation. Our findings allowed us to develop a tool for performing automated deception detection. Due to its technical simplicity, the detection algorithm can be easily integrated into reviewing platforms or other programs. The findings also allowed us to develop guidelines for helping individual consumers detect fake reviews. However, we found that additional information does not improve consumers' low accuracy in fake review detection. Because of the human tendency to over-evaluate the veracity of information (i.e., the veracity effect) and the presence of fake online reviews on reviewing websites, we prove it necessary to protect consumers with external tools and procedures. Overall, we contribute to the growing, but still underdeveloped field of cyber deception research (Stech, Heckman, Hilliard, & Ballo, 2011).

2. Theoretical background and hypothesis development

2.1. Fake online review detection

It is important to take fake online reviews seriously (Malbon, 2013). Multiple studies show that consumers rely heavily on consumer reviews when making purchase decisions (e.g., Nielsen, 2017; Plotkina & Munzel, 2016). Online reviews written and published by peer consumers offer a mechanism for regulating the marketplace by pointing out products and services of inferior quality (Malbon, 2013). Genuine consumer reviews can moderate bad seller behavior and improve the quality and efficiency of the marketplace. As marketers are increasingly aware of the effect of online reviews on consumers, reviews on products and services have become an important element of the marketing strategy (Chen & Xie, 2008). Unfortunately, some companies have responded to this shift of control from marketing-induced to consumer-induced communication by generating fake online reviews. In so doing, such firms undermine a (potentially) effective and efficient mechanism for overcoming information asymmetry between online sellers and buyers, as fake reviews represent an important threat to consumers' ability to make better-informed purchasing decisions, potentially decreasing their overall well-being (Lappas, 2012).

Although policymakers and regulators have started to actively address the issue of fake reviews (e.g., the updated guidance by the International Consumer Protection and Enforcement Network[1]), legal actions against deceivers are complex and difficult due to the lack of detection and identification of perpetrators. Therefore, the main concerned parties (i.e., consumers and reviewing platforms) should also consider taking active steps to filter out deceptive online reviews. To do so, both reviewing platforms and individuals should be able to detect opinion spam.

The main approaches to opinion spam detection are based on the linguistic characteristics of reviews (NLP or Natural Language Processing), the characteristics of the reviewer, or some combination of the two (see Table 1). Researchers agree that there are no distinct words that can help to distinguish a fake review from a truthful one (Crawford, Khoshgoftaar, Prusa, Richter, & Al Najada, 2015). Therefore, the most common approach is to use a set of micro-linguistic characteristics. For instance, Jindal et al. (2010), Li, Huang, Yang, and Zhu (2011), and Mukherjee, Liu, and Glance (2012) used n-grams (individual or groups of words used as features), while Shojaee et al. (2013) used syntactic and lexical features to separate fake from truthful reviews. To obtain better detection algorithm performance in terms of the probability of correctly detecting deceptive reviews, existing research suggests using a mix of features available in the text and context of the review.

The literature on the detection of fake online reviews emerged only recently, illustrating the ongoing quest for the most effective linguistic analysis software, labeled and representative dataset, and performant classifier. Differences in method design lead to contradictory results concerning deception cues. For instance, Yoo and Gretzel (2009) found that linguistic complexity is higher in deceptive reviews, while Harris (2012) found the opposite. In these circumstances, we chose not to hypothesize on the exact deception cues to be used but, rather, to explore the cues that might help distinguish truthful messages from fake ones. Such an exploratory approach is also applicable because we use a linguistic analysis software that has not been used in prior studies on fake online reviews (i.e., Coh-Metrix, Graesser, McNamara, & Kulikowich, 2011). Finally, as micro-linguistic cues should help to separate deceptive from truthful reviews, we specifically expected that a certain combination of linguistic cues, rather than stand-alone linguistic cues, would be more effective for fake review detection (Fuller, Biros, & Delen, 2011). Our first hypothesis is as follows:

H1. Compared to a single cue algorithm, a detection algorithm that combines multiple micro-linguistic cues detect deceptive online reviews with higher accuracy.

Yet, existent research insists that positive and negative reviews should be addressed separately (Ott et al., 2011; Ott, Cardie, & Hancock, 2013), which usually requires different algorithms to match the valence of the reviews. This assertion is rooted in psycholinguistics, which stipulates the effect of emotion on deception cues. Obviously, faking positive and negative accounts impacts individuals in different ways, which may lead to different sets of deception cues (e.g., Ekman, 1985).

The emotional dimension of the reviewer might also be impacted by his/her motivation to write a fake review. Hurkens and Kartik (2009) have shown that the probability that a person will choose to deceive depends on the incentive for the deception. We assume that, in real life, deceivers have two main motivations: to post reviews on their own businesses (reputation or social motivation) or to be paid to do so by another business (monetary motivation). DePaulo et al. (2003) have

[1] https://www.icpen.org/initiatives.

Table 1
Summary of existing research on fake online detection (adopted from Crawford et al. (2015) and completed).

Research	Dataset	Features used	Detection algorithm	Performance metric	Score	Method complexity
Jindal and Liu (2007)	5.8 million reviews from Amazon	Review and reviewer	LR	AUC	78%	Low
Jindal and Liu (2008)	5.8 million reviews from Amazon	Review, reviewer and product	LR	AUC	78%	Medium
		Text features	LR	AUC	63%	Low
Yoo and Gretzel (2009)	82 reviews written by marketing students	Quantity, complexity, diversity, immediacy, branding, sentiment	*t*-Tests	P-value	NA	Low
Jindal, Liu, and Lim (2010)	6000 reviews from Opinions	Review and reviewer	NV	F-Score	0.631	High
Ott et al. (2011)	800 hotel reviews from TripAdvisor (true) and AMT (fake)	Bigrams	SVM	Accuracy	89.6%	Low
Lau et al. (2011)	Dataset collected from Amazon	Text and style features	SLM	AUC	0.9986	High
		LIWC + Bigrams	SVM	Accuracy	89.8%	Medium
Harris (2012)	400 truthful reviews from websites, 400 fake reviews crowdsourced	Yoo and Gretzel (2009) + polarity and readability	SVM	Accuracy	73.8–76.3%	Medium
Mukherjee, Venkataraman, Liu, and Glance (2013)	Yelp's real-life data	Behavioral + bigram	SVM	Accuracy	86.1%	Medium
Shojaee, Murad, Bin Azman, Sharef, and Nadali (2013)	Ott et al. (2011)	Stylometric features	SVM	F-measure	84%	Low
Ott et al. (2013)	Ott et al. (2011)	n-Gram features	SVM	Accuracy	86%	Low
Feng and Hirst (2013)	Ott et al. (2011) and product profile	n-Gram and syntactics	SVM	Accuracy	90.1%	Medium
Hernández, Guzmán, Gomez, and Rosso (2013)	Ott et al. (2011)	n-gram	SVM, NV	F-measure	0.837	Medium
Li, Ott, and Cardie (2013)	Ott et al. (2011) + 400 fake reviews	LIWC + POS + Unigram	SAGM	Accuracy	65%	High

LR: Logistic Regression; AUC: Area Under the ROC Curve; NV: Naïve Bayes; SVM: Support Vector Machine; SAGM: Sparse Additive Generative Model; SLM: Semantic Language Model.

found that deception cues are more pronounced when people are motivated to succeed, especially when the motivation involves reputation rather than money. We therefore expect that the valence of the fake message and the motivation to deceive will impact the linguistic cues of deception:

H2a. Valence moderates linguistic features, such that micro-linguistic cues that help to separate fake online reviews from truthful ones are different for positive and negative reviews.

H2b. The reviewer's motivation to deceive moderates linguistic features, such that micro-linguistic cues that help to separate fake online reviews from truthful ones are different for a socially motivated (i.e., reputation) reviewer and a monetary motivated reviewer.

2.2. Human deception detection

Individuals seem to be vulnerable to online fraud despite fairly strong indicators, such as fiction seals, warranties, and news stories (Grazioli & Jaryenpaa, 2000). Based on the findings of previous research, we might conclude that deception cues exist overall (e.g., Ott et al., 2011; Zhou & Zhang, 2008), but that they are either difficult for humans to detect (Bond & DePaulo, 2006) or perceived and used incorrectly (Riggio & Friedman, 1983).

Prior studies on detection deception in transcripts of accounts (e.g., Landry & Brigham, 1992) or written statements (Toma & Hancock, 2012) show very limited deception detection accuracy for human detectors. Vrij (2000) found that the average accuracy of human lie detectors was only slightly above the level of chance (56.6%). This supports similar findings by Kraut and Poe (1980) (approximately 57%). Thus, we expect that individuals are inaccurate at deception detection and hypothesize the following:

H3. Compared to the automated algorithm, consumers detect deceptive online reviews with lower accuracy.

Consumers have become increasingly suspicious toward online reviews (Larson & Denton, 2014). However, in general, individuals are not suspicious when it comes to deception: they routinely judge lies as truth. This trusting tendency is called the truth bias or veracity effect (Levine, 2014; Levine et al., 1999). The truth bias is the cornerstone of interpersonal deception theory (IDT) (Buller & Burgoon, 1996). It explains how individuals handle actual and perceived deception. Namely, IDT states that individuals give away deception cues through their verbal and non-verbal communication, but that most people misperceive or do not notice these cues, evaluating a considerable amount of deceit as truth. The truth bias has been supported for most conditions and deceit situations, and there are strong grounds to believe that the same holds for online reviews (Bond & DePaulo, 2006; Vrij, 2000). We therefore hypothesize that:

H4. Consumers are more likely to classify fake reviews as truthful than to classify true reviews as fake.

The truth default theory (Levine, 2014) focuses on the veracity effect and stipulates that truth bias is not a cognitive flaw, but is functional, adaptive, and conducive to correcting inference. Therefore, deception detection accuracy is a linear function of truth–lie base rates and can be improved. External and personal factors can affect accuracy and the deception detection process. For instance, initial beliefs about the veracity of information can either reinforce the existing truth bias or stimulate suspicion (Granhag & Stromwall, 2000). The main deception-related theories—for example, IDT (Buller & Burgoon, 1996), persuasion knowledge theory (Friestad & Wright, 1994), and deception detection with multiple senders (Marett & George, 2004)—consider the target's knowledge of the context and deception practices to be important factors. Vrij, Granhag, and Porter (2010) show that people often fail to detect lies because of: (1) common misperceptions of deception cues, (2) sophisticated deceits and complex contexts, or (3) a lack of motivation. As the context of deceit in the present study (an online platform with online reviews on restaurants) is predefined and rather well known to an average consumer, we focus on consumers' knowledge of cues to detect fake online reviews and their motivations. Consumers often see alerts on the presence and dangerous impact of online reviews in mass media[2] and on the websites of legislators, watchdog organizations,[3] and even reviewing platforms.[4] These messages usually include tips on how to detect opinion spam; however, their actual impact is unknown. At the same time, consumers are increasingly exposed to a multitude of information quality labels, such as "trusted reviews", "verified purchase", and similar. We believe that advice on how to detect fake online reviews motivates consumers to be more vigilant and provides them with a tool to better identify the nature of online reviews, while quality labels for trusted reviews reduce motivation to detect online spam. We therefore expect that:

H5. Information on the presence/absence of fake online reviews will increase/decrease consumers' accuracy in detecting fake online reviews.

3. Study 1: analysis of an algorithmic detection of fake online reviews

3.1. Purpose and experimental approach

Researchers on deception have argued that the future of deception detection lies in the analysis of verbal, rather than non-verbal, cues (Vrij, 2008). Verbal deception cues have increased in importance because of the digitalization of communication and advances in analysis software. An important question, however, is which linguistic samples should be used to establish deception and truth linguistic cues. The absence of fully labeled samples of truthful and fake reviews is considered one of the biggest challenges of online spam detection (Crawford et al., 2015). Many studies on fake reviews crawl large samples of reviews and cluster them to find patterns and features that might reveal fakery (e.g., Feng, Xing, Gogar, & Choi, 2012; Jindal & Liu, 2007, 2008).

When the truthfulness of reviews is known, they become reliable samples for linguistic analysis, also called "gold standard corpora" (Gokhman, Hancock, Prabhu, Ott, & Cardie, 2012). One particularly convenient and valuable way of collecting deceptive and truthful reviews to establish this kind of corpora is crowdsourcing. Crowdsourcing platforms connect people who need numerous small tasks performed and workers who are ready to carry out these tasks. The advantage of crowdsourcing lies in the similarity of this context to the real-life situation of online reviewing. Mihalcea and Strapparava (2009) were among the first to use Amazon Mechanical Turk (AMT) crowdsourcing platform to collect deceptive and truthful opinions. Ott et al. (2011) also used AMT to produce gold-standard deceptive content. Because previous studies have shown the advantages of creating deception gold standard corpora through crowdsourcing (Gokhman et al., 2012), we used this method to create our corpora as well. However, existing studies chose to crowdsource fake online reviews and to assume that highly ranked reviews available online were a good proxy for truthful reviews (e.g., Ott et al., 2011). To ensure a fully labeled gold corpora of reviews, we crowdsourced both truthful and deceitful reviews.

3.2. Data collection: measures and participants

We followed the baseline of gold-standard corpora collection (Gokhman et al., 2012). We collected a set of 1041 truthful and deceptive reviews in the context of a restaurant service. All the reviews were generated under the same guidelines. Respondents were asked to write a review on a familiar service—a restaurant or a cafeteria. We asked respondents to compose either a true online review or a fake online review. In the case of a fake review, respondents were invited to submit either a positive or a negative fake review. For the fake review scenario, the respondents were asked to simply write a fake review (no stimuli), to write a fake review as if they worked in the restaurant (social stimuli), or to write a review to receive higher compensation on the crowdsourcing platform (monetary stimuli). We controlled for socio-demographics among the samples (e.g., gender (Dreber & Johannesson, 2008) and age (Caspi & Gorsky, 2006)) because prior research has shown that these parameters influence customers' behaviors when they are asked to deceive.

A pre-test was conducted prior to launching the main research. A group of 28 graduate students completed questionnaires, commenting on their understanding and perceptions as part of their written answers to the questionnaires and during a subsequent group discussion. The pre-test resulted in minor modifications to the wording.

The respondents were recruited via AMT. To participate in our study, AMT workers had to be "masters" with at least an 85% Human Intelligence Task (HIT) approval rate. Participants were assigned randomly to one of the various experimental conditions. The 1327 completed responses initially gathered were examined in three steps: (1) correct use of the code respondents had to paste to AMT to obtain payment, which was placed randomly in the middle of the questionnaire and changed daily, (2) the correct answer to an attention check question, and (3) manual verification of the quality of the review. The attention check was adopted from Paolacci, Chandler, and Ipeirotis (2010) and stated: "While watching the television, have you ever had a fatal heart attack?" The retained reviews were of the demanded valence, did not contain swear words, and were relevant to the topic given in the scenario (i.e., a restaurant or a café). Respondents also responded on a two-item realism scale adopted from Freberg (2012) to the following two questions: (1) "The situation described was realistic" and (2) "I had no difficulty imagining myself in the situation." The mean of the realism score was 5.14 (on a scale of 1 – low to 7 – high) and was judged acceptable for proceeding with the data analysis (Dabholkar, 1994).

The final sample consisted of 1041 respondents: 206 in Condition 1 (true review) and 835 in Condition 2 (fake reviews). Of the 835 respondents who submitted a fake review, 392 were asked to write a positive review, and 443 were asked to write a negative review. The randomization allowed comparably equal samples for every stimulus (social, monetary, or none). To obtain equal sample sizes, an additional 87 truthful reviews were collected via personal social networks. These additional reviews did not significantly differ in characteristics from the reviews obtained through AMT. During the algorithm development phase, the complete sample was divided into a training data set (70% of complete data set) and a test data set (30% of complete data set). The distribution of the responses among the conditions is shown in Table 2.

The respondents comprised 54% men and were, on average, 26.3 years old (range: 19 to 65 years). A vast majority of respondents held a bachelor's degree (54%) or a master's degree (23%). For 89%, English was their mother tongue or second language with perfect knowledge; 56% consulted online reviews regularly, while 18% did so very often; and 34% posted online reviews on a regular basis.

3.3. Data analysis

Manual labeling is time-consuming and not efficient (Crawford et al., 2015). The attractiveness of automated text analysis lies in its

[2] https://www.nbcnews.com/business/consumer/fake-online-reviews-here-are-some-tips-detecting-them-n447681.

[3] https://www.smh.com.au/technology/watchdog-eyes-fake-online-reviews-20131203-2yo7a.html.

[4] https://www.searchenginepeople.com/blog/yelp-consumer-alert-for-fake-reviews.html.

Table 2
Distribution of the respondents in Study 1 among the conditions.

Condition	Valence	Motivation	Training set 70%	Testing set 30%	Total
Fake	Negative	No stimuli	100	47	147
		Social stimuli	100	47	147
		Monetary stimuli	100	49	149
	Positive	No stimuli	100	30	130
		Social stimuli	100	31	131
		Monetary stimuli	100	31	131
True	Negative		100	43	143
	Positive		100	50	150
Total			**800**	**328**	**1128**

higher reliability, greater detail, and faster speed than manual human analysis (Graesser et al., 2011). The cue-based analysis was carried out with the help of two linguistic analysis tools: Language Inquiry and Word Count (LIWC) and Coh-Metrix. By adopting these two analysis tools, the present study compares a word-oriented and dictionary-based linguistic analysis (LIWC) with a semantic-syntactic approach (Coh-Metrix).

LIWC is a transparent text analysis program that counts words in psychologically meaningful categories (Tausczik & Pennebaker, 2010). LIWC is widely applied in social sciences: for instance, to detect personality traits (Mairesse, Walker, Mehl, & Moore, 2007), to predict academic performance (Robinson, Navea, & Ickes, 2013), and, of course, to analyze deception (Hancock, Curry, Goorha, & Woodworth, 2007; Mihalcea & Strapparava, 2009; Vrij, Mann, Kristen, & Fisher, 2007). LIWC has already been successfully used to detect fake online reviews (Ott et al., 2011). The LIWC analysis contains nearly 90 scales in the following dimensions: (1) self-referencing, (2) social words, (3) positive emotions, (4) negative emotions, (5) cognitive words, (6) articles, and (7) long words (more than six letters). The LIWC process relies on an internal dictionary that defines which words should be counted in every dimension within the analyzed text files. One word is processes at a time: the program searches for a match with words in the dictionary and tags every word according to the appropriate word category scale. The default LIWC Dictionary (the latest LIWC2015 version) comprises nearly 6400 words, word stems, and selected emoticons. Each word can belong to more than one scale.

Coh-Metrix analyzes texts based on multiple measures of language and discourse (Graesser et al., 2011). This software characterizes texts at multiple levels of language and discourse (Graesser et al., 2014). Usually, Coh-Metrix is used to analyze writing quality (e.g., Crossley & McNamara, 2011). It has previously been combined with LIWC to investigate gender differences in linguistic styles, yielding more detailed results (Bell, McCarthy, & McNamara, 2012). The final output allows 101 linguistic features characterizing every text. These measures are crosslinked to theoretical frameworks from psychology and psycholinguistics. The text indices are divided into five major groups that characterize: (1) word concreteness, (2) syntactic simplicity, (3) referential cohesion, (4) causal cohesion, and (5) narrativity. Cohesion relates to the structure of the text, which can help the reader mentally connect ideas in the text (Graesser et al., 2011), while narrativity is familiar language, as in an everyday conversation. Thus, the tool allows for analyzing the text from quantitative and qualitative perspectives and processing the text both word-by-word and as a whole.

3.4. Results and discussion

We carried out all the initial analyses on the training set, which consisted of 440 positive and 440 negative reviews. Then, we tested the predictive accuracy on the testing sets of each valence separately before we carried out an ANOVA in SPSS with linguistic cues obtained from the LIWC and Coh-Metrix analysis tools for every linguistic feature provided by this software (Appendix A). The results obtained through LIWC showed only one significant difference valid across positive and negative reviews: true messages were more emotional than fake messages. All other LIWC measures were opposite for positive and negative reviews. This might be concerning, as the valence of a review is sometimes hard to define, especially when it is analytical (e.g., "The restaurant offers high quality food, but is pricey"). We know that LIWC output depends largely on context, as it is a dictionary-based software and is, therefore, difficult to adapt to every situation. At the same time, the reviews written for different motivations differed only in terms of the use of personal pronouns ("we") and conjunctions.

The Coh-Metrix output gave more general results. We found that true messages were generally longer (number of paragraphs and of sentences per paragraph), easier to read, and more precise in terms of wording (hypernymy); however, they lacked temporal cohesiveness. Furthermore, true messages contained more connectives (logical and adversative) and less adjectives. Coh-Metrix confirmed that positive and negative online reviews have different linguistic structures; however, while the reviews of different valences did not share all of the same cues, they also did not contradict one another. Based on this evidence, we decided to analyze positive and negative reviews separately. The comparison motivations to write a fake review showed significant differences for only one feature: syntactic simplicity, which was lower for the control group than for the reputational review (social motivation) group. This feature was excluded from further results.

None of the aforementioned cues offered a satisfactory level of detection when taken alone ($< 25\%$). Therefore, we performed binominal logistic regressions with all the linguistic features provided by every software. The binominal logistic regression was chosen due to its simplicity and overall good performance compared to more sophisticated classifiers (Crawford et al., 2015). Furthermore, previous research has found that it has the lowest false positive rate, potentially making it preferable for deceit detection (Abu-Nimeh, Nappa, Wang, & Nair, 2007).

First, we ran the binominal regression for every type of fake review of each valence (monetary motivation, social motivation, and none), comparing them with the authentic reviews. We included all the cues available for each software using the Wald ascending procedure, so that only the cues that were significant on their own and that contributed to the overall accuracy of the regression were retained. We then merged all the linguistic features that were significant in every sub-test and had the same valence: this procedure aimed to identify fake reviews regardless of their underlying motivation. For both linguistic programs, the caption cues identified for every subset of motivation were coherent in terms of valence and coefficient size. The highest absolute value of every coefficient was retained and merged into a common algorithm for positive and negative reviews. The final algorithms were run on the test sets.

The LIWC output performed moderately on the training set (i.e., positive deception detection, negative deception detection, positive true detection, negative true detection) and was even less conclusive on the test set (i.e. overall detection between 52% and 58%). The deception cues were again contradictory for positive and negative reviews and could not be merged into one generalized algorithm. Therefore, we ruled out LIWC as an applicable tool for fake online review detection with a binominal logarithmic regression.

Binominal regressions provide more satisfactory detection accuracy when trained and tested with Coh-Metrix output. Thus, for the positive reviews, the algorithm identified 82.4% of fake reviews and 66% of true reviews, with an overall detection rate of 76.6% (Table 3). For the negative reviews, the detection accuracy was 66.8%, with 53.4% of true reviews and 83.9% of fake reviews identified correctly. We also found that deception cues were not contradictory for positive and negative reviews, and some were even close in value. Therefore, the algorithms to detect positive and negative reviews could be merged into one. We identified a total of 29 indices that, once combined into a binominal

Table 3
Results of automated deception detection.

Sub-set		Detection by an algorithm	Accuracy of detection as percentage	Overall detection accuracy
Training set: Positive	True	75/110	68.2%	80.9%
	False	103/110	93.6%	
Testing set: Positive	True	33/50	66%	76.6%
	False	75/91	82.4%	
Training set: Negative	True	71/110	64.8%	77.8%
	False	100/110	90.9%	
Testing set: Negative	True	23/43	53.4%	66.8%
	False	120/143	83.9%	
Merged algorithm (training set positive and negative)	True	54/93	58%	81%
	Fake	210/234	89.7%	

logarithmic regression, yielded 81% detection accuracy, with 89.7% deception detection accuracy and 58% truth detection accuracy. However, while the merged algorithm improved the detection of fake reviews, it had a rather high false negative error (42%). Therefore, it might be preferable to exploit the algorithm for positive reviews separately and to work on improving the accuracy of the negative opinion spam detection.

We also tested our different hypotheses. Hypothesis 1 is valid: fake online reviews and genuine truthful reviews can be distinguished based on a set of linguistic cues. The detection algorithm is satisfactory compared to previous approaches. While its accuracy is higher than those in some previous works (e.g., Jindal and Liu (2008): 63%; Li et al. (2013): 65%; Harris (2012): 76.3%; Feng, Xing, et al. (2012): 65.5%), it does not attain the overall accuracy of 86% achieved by Ott et al. (2011). Most importantly, the results of our analysis show that positive and negative reviews have different linguistic structures (Hypothesis 2a), but that a combined detection algorithm based on semantic and syntactic features can detect fake reviews of both valences with sufficiently good accuracy, compared to previous online spam detection studies. Furthermore, the detection algorithm can identify reviews written by people with different motivations to deceive (Hypothesis 2b). The Brier scores and the Receiver Operating Characteristic (ROC) curve are presented in the Appendix B. We support the applicability of the binominal regression as a classifier that is very easy to perform and integrate into any web application or platform. We compared the detection accuracy of the bi-log regression with that of random forests (Appendix C). While the cues used by the random forests are similar, the detection accuracy is slightly lower than that achieved with the bi-log regression, and the processing is more consuming. Discriminant regression might be a good way to classify online reviews (Zhou, Burgoon, Zhang, & Nunamaker, 2004); however, it is not applicable due to the high collinearity of the data in the Coh-Metrix tool (e.g., number of words in a phrase and total number of words). We believe that SVM classifiers have high potential; however, they would be more appropriate on a bigger sample (Crawford et al., 2015).

The deception cues analysis provided deeper insight into what constitutes a fake online review and what could be used as a simple automated tool or guideline for spotting fake reviews (Appendix D). Broadly, we can conclude that fake messages are well written, as if following a specific scenario (e.g., coefficient of temporal cohesion = 312.282, p-value < .1; coefficient of concrete words = 128.953, p-value < .1; for example, "We often go to this restaurant and I always [*temporal cohesion*] opt for the chef's cassoulet [*concrete words*]"). Furthermore, fake messages lack temporal references and structure (coefficient of time preciseness = −0.019, p-value < .1; coefficient of temporal connectives incidence = −0.502, p-value < .1; for example, fake: "the food is great in this place;" authentic: "we went to this restaurant after visiting the Christmas market" [*time preciseness and connectives*]). Finally, fake messages contain more adjectives than verbs, or more description than action (coefficient of adjective incidence = 0.589, p-value < .1; coefficient of intentional verbs = −0.494, p-value < .1; for example, "the pizza is great here, juicy and fresh!" [*syntactic pattern density*]).

These findings correspond with those of some previous studies, while contradicting those of others. Cognitive complexity was, indeed, lower in deceptive messages (Newman, Pennebaker, Berry, & Richards, 2003), but the structure of the messages was good. Furthermore, descriptions (adjectives and adverbs) were more prevalent than actions (verbs). This is intuitive and similar to the results of some other studies (Toma & Hancock, 2012). Contradictions in research findings on deception detection are quite common (e.g., Ott et al., 2011), as studies seldom give precise indications of all the cues used (Zhou et al., 2004).

Next, we sought to determine whether the findings on the features shared by fake reviews could be used to inform and protect consumers. More precisely, we sought to answer our main research question: Does an automated detection tool really outperform human judges in deception detection? To do so, we conducted a second experiment.

4. Study 2: experiment on human detection of fake online reviews

4.1. Purpose and experimental approach

Based on the findings of Study 1, the second study focused on consumers' ability to correctly identify online reviews as truthful or fake. We asked respondents to evaluate the trustworthiness of online reviews. Prior empirical research has revealed important limitations to individuals' capacities to separate true from fake online reviews (Crawford et al., 2015). Furthermore, research conducted by Levine et al. (1999), and Levine (2014) identified a truth bias, which is the tendency for (human) message judges to infer honesty independent from actual message veracity. Therefore, Study 2 aimed to experimentally examine the presence of a potential truth bias in the human detection of fake online reviews.

4.2. Data collection: measures and participants

The respondents were recruited again via AMT with the same participation conditions as in the first study (i.e., masters workers with at least an 85% HIT approval rate, a changing validation code position within the questionnaire, a question for attention check, and a quality check of responses). The final sample consisted of 407 respondents. First, we evaluated the respondents' experiences with online reviews and the consumption context, suspiciousness toward online reviews, cognitive capacities, need for cognition, and imagination. We then exposed the respondents to a scenario in which they were consulting a reviewing platform to choose a restaurant for a lunch break and came across a certain restaurant with 10 online reviews. The reviews were equally distributed between true and fake and between positive and negative, and they were put in a random order to avoid order bias. The reviews were taken from Study 1 and were chosen from among the reviews identified correctly by the automated algorithm. Respondents had to evaluate every review as either true or fake (binary option). The respondents were randomly distributed into three equal experimental conditions: (1) the first group received no additional information; (2) the second group received an alert sign and a guideline on how to spot a fake review based on the results of Study 1; (3) the third group received a quality label "trusted reviews" attached to all the reviews to which they were exposed. We used SPSS software to evaluate detection accuracy to test the effect of the control variables on detection rates.

The respondents comprised 56% men and 44% women and were between 20 and 59 years old, with an average age of 32.4 years. A vast majority of the respondents were educated with either a bachelor's degree (49%) or a master's degree (21%). For 93%, English was their mother tongue or second language with perfect knowledge, and 61%

consulted online reviews regularly, while 16.5% did so very often. The personal and psycho-demographic characteristics of the respondents were controlled so that they were comparable among the three experimental conditions.

As deception detection can be impacted by knowledge of the communication context, the discussed subject, and deception in general (Friestad & Wright, 1994), we control for these variables. The measures were adapted from previous studies. The respondents answered on 7-point Likert scales from 1 (“totally disagree”) to 7 (“totally agree”). Measures for general online review knowledge were adapted from Chu and Kim (2011). Experience in consulting online reviews on restaurants was adapted from Martin and Stewart (2001). The scale for suspicion toward online reviews was taken from Obermiller and Spangenberg (1998): this scale is focused solely on the truthfulness of the online reviews and was found to be more applicable to the experimental context than other scales that consider reviewers' motivations and identities (Zhang, Ko, & Carpenter, 2016). The reliability of the scales was revalidated and found to conform to general research requirements, with satisfactory inter-item correlation and a Cronbach's α in the range of 0.7 to 0.853 (Black, Hair, Babin, & Anderson, 2009). The realism scale was adopted from Freberg (2012). The mean of the two-item realism score was 5.08 and was judged acceptable for proceeding with the data analysis (Dabholkar, 1994).

4.3. Results and discussion

The aggregated results (cf. Table 4) show that, in all three groups, the respondents had an above-chance level of overall detection accuracy ($D_{total} = 57\%$). This rate is significantly different from the chance level ($p < .001$), but is still not very high. There was no significant difference in overall detection accuracy among the experimental conditions.

When analyzing the division between truth and deceit detection, we found that respondents detected truth much better than they detected deceit (no information: $D_{truth} = 69\%$ vs. $D_{deceit} = 45\%$; alert: $D_{truth} = 63\%$ vs. $D_{deceit} = 52\%$; quality label: $D_{truth} = 71\%$ vs. $D_{deceit} = 45\%$). There was a significant difference ($p < .001$) in detection rates of truth and deceit in every group. Furthermore, deceit detection in the control group with no information and in the group under the trust label condition was significantly below the chance level ($D_{chance} = 50\%$). The alerted consumers, however, showed no evidence of deceit detection significantly higher than the chance level. At the same time, truth detection rates were significantly different among the conditions. Indeed, we found that alerting and educating consumers about how to detect fake reviews does not help them become dramatically better deceit detectors; rather, it merely makes them more suspicious, which lowers their truth detection. The opposite is true when the consumers are exposed to quality labels: these consumers have a higher truth bias.

This led us to conclude that people are not good at detecting fake online reviews: more than half of fake online reviews remain undetected. This result supports Hypothesis 3: consumers under-perform all reported automated tools in fake review detection tasks. We also found evidence supporting Hypothesis 4 (i.e., fake reviews are judged as genuine reviews more often than true reviews are judged as fake reviews). Finally, we found that situational information indeed impacts consumers' detection accuracy (Hypothesis 5). While the trusted reviews label comforted the respondents, it also improved their detection of truthful reviews. Information on the presence of fake online reviews decreased consumers' truth bias and truth detection accuracy, but did not significantly improve their deceit detection.

In Study 1, we found that fake online reviews can be efficiently detected by an automated algorithm. However, in Study 2, we showed that consumers, even when primed with tips on deception detection, fail to distinguish between genuine and fake reviews. The remaining question is whether fake reviews represent a real danger and whether they are frequent on reviewing websites. To answer this last question, we carried out a field study, which is described in the following section.

5. Study 3: application of the algorithm in the field study

5.1. Purpose and approach

To gain some perspective on the results obtained in Studies 1 and 2, Study 3 put the algorithm developed in Study 1 into practice. In the algorithm field validation process, we downloaded reviews on the five best-rated and three worst-rated restaurants in a European city from three widely used reviewing websites: TripAdvisor, Yelp, and La Fourchette (in English: The Fork). All reviews were downloaded manually from July 1 to 15, 2015. The complete sample contained 3891 reviews: 1595 from TripAdvisor, 780 from Yelp, and 1516 from La Fourchette. These reviews were further processed and analyzed using the same online linguistic tool we used during the first experiment: Coh-Metrix (tool.cohmetrix.com) (Duran, Hall, McCarthy, & McNamara, 2010). The resulting data (positive and negative reviews) were then analyzed with the combined algorithm obtained from the first study. To calculate the probability of every review being fake, we used a function that was the reverse of the initial binominal logarithmic regression. SPSS uses the natural log of the variable; therefore, an exponential function can be used to convert natural-logged forecasts. The following standard inverse function calculated the probability of every review being fake, taking into consideration every deception cue that contributes to the deceit detection (*index*) according to its *coefficient*:

$$P(fake) = \frac{1}{1 + e^{-coefficient*index}}.$$

5.2. Results

The exponential function showed values infinitely close to 1 (probability of being fake) or infinitely small and, therefore, negligible (high probability of being true). Thus, the standard threshold of deceptiveness (if $P(fake) > 50\%$) was judged sufficient. The percentages of reviews that were most likely fake according to our algorithm were as follows: 11.97% on TripAdvisor, 14.14% on Yelp, and 11.4% on La Fourchette.

All the reviews identified as potentially fake were manually analyzed by three independent human judges. The manual analysis aimed

Table 4
Results of human deception detection.

Experimental condition	Truth detection	Deceit detection	Total detection	Within group comparison deceit/truth	Between group comparison truth	Between group comparison deceit
Control group: no information	69%	43%	57%	$\alpha < 0.001$	$D.Control_{truth} > D.Alert_{truth}$ $\alpha < 0.01$	$D.Control_{deceit} < 50\%$ $\alpha < 0.001$
Alert and detection guideline	63%	52%	58%	$\alpha < 0.001$	$D.Alert_{truth} </ D.Label_{truth}$ $\alpha < 0.01$	
Trusted reviews quality label	71%	45%	57%	$\alpha < 0.001$	$D.Label_{truth} > D.Alert_{truth}$ $\alpha < 0.01$	$D.Label_{deceit} < 50\%$ $\alpha < 0.01$

to understand whether the reviews that were suspicious to the automated algorithm were equally suspicious to a consumer. Furthermore, this procedure helped us gain insights into the contextual features of the reviews identified by the algorithm as untrustworthy. Indeed, the nature of the reviews suspected to be fake differed on every site. This might have been the result of different website functionalities and reviewer communities. In the following, we discuss the characteristics of the fake reviews on every online platform.

5.3. Discussion

Study 3 highlights the functioning of three reviewing websites. On TripAdvisor, emphasis is placed on reviewing and community management. Reviewers are given scores, badges (e.g., senior reviewer), and levels (based on scored points from one to six in increasing importance). This system encourages reviewers to be active by writing, responding, posting, voting, and posting photographs. All profiles are public and easily viewed. Therefore, there is an apparent transparency and ease of use that might communicate more trustworthiness for reviews with high scores posted by alleged reviewers. The 191 reviews identified by the algorithm as potentially fake fell into two categories: those deemed very suspicious (40%) and those posted by a reviewer with a seemingly trustworthy profile (60%). The detection results for the reviews identified as potentially fake were very enlightening. We found that some reviewers on TripAdvisor used the same text for reviews on different websites, for the same restaurant but using different profiles, and for reviews of different restaurants. Previous work on online spam points to duplicate content as a potential sign of fakery (Jindal & Liu, 2008). In addition, among the reviews that looked suspicious were reviews by profiles with little to no personal information: just a single or a few reviews posted at once and a long time ago. These messages were of short and medium length and were quite general (e.g., “good food,” “good service,” “we were not disappointed,” or “will certainly return to this place”).

The second sample of 115 reviews presented a bigger problem because, at first glance, the reviews were good, but the algorithm was wrong. However, we identified three interesting characteristics that united these “almost good” reviews. First, they all reviewed the top-ranked restaurants. Second, they all included very exact details on the menu (specific dishes, with details on the preparation and presentation), the waiters, and the owners (including their names and personal relationships). They even included justifications for the high price of such culinary experiences. Third, each message included a call to action, with a voiced recommendation and a rather detailed explanation of how to proceed with reservations and orders. These messages were usually long and contained personal information and unnecessary observations (e.g., “it was our anniversary,” “we took a taxi,” “it was a nice chilly September evening”) and an emotional touch. After a careful reading of messages in this sample, we allow that some of them were possibly genuine. However, a considerable number of reviews for the same restaurant used the same structure, as if the reviewers had been given a questionnaire to complete. These visible patterns led us to doubt the reviews and to question whether they were sponsored reviews from customers with good profiles.

On Yelp, there was much less visibility and information about the reviewers, as the company has merged with two other online platforms: Cityvox and Qype. Thus, there was little to no information on the messages coming from these sources. Surprisingly, most of the suspicious reviews belonged to these untraceable reviewers. The reviews that were detected as possibly fake were of short and middle length and frequently mentioned the reasonable prices, the atmosphere, and general information (e.g., the menu or hours of operation). Within these reviews, some were from the Yelp Elite: individuals identified as opinion leaders and active members of the Yelp community. This indicated the inaccuracy of the algorithm, but was easily explained by the nature of the reviews. The messages posted by some of the Elite members bore more resemblance to those of personal blogs (habits, general opinions, etc.) than reviews of a restaurant. The percentage of potentially deceptive reviews on Yelp obtained through the algorithm developed from insights in Study 1 (i.e., 14.14%) was similar to those found in recent empirical studies. Luca and Zervas (2016) reported that about 16% of the Yelp reviews in their sample were identified as being suspicious and were, therefore, filtered by the review site's team.

The website La Fourchette had the least community management and the least information on the reviewers. The profiles were not personal, and the reviews were rather short. It was difficult to evaluate the trustworthiness of the reviews, but upon comparison of “detected” and “not detected,” it was evident that the former contained more opinions than descriptions and used more superlatives. In general, the reviews that were deemed suspicious according to our detection algorithm were mostly positive. They were also more extreme or enthusiastic than those identified as potentially genuine.

6. General discussion

Based on our results, we can draw four main conclusions: (1) True and fake online reviews have significant micro-linguistic differences that can be used to detect their nature. (2) People are not good at detecting fake online reviews. (3) Even the biggest reviewing websites include potentially manipulative reviews. (4) An automated detection tool is a necessity because it has a real deception detection capacity. Thus, this research makes several academic and managerial contributions.

6.1. Research implications

First, we extend the analysis of the psycholinguistic deception cues of online reviews (Zhou et al., 2004). Second, our findings allow us to both propose guidelines to help consumers better evaluate the trustworthiness of online reviews and develop an automated deception detection tool. The latter may be used by reviewing websites to improve the quality of reviews posted on their sites, by consumers to support their purchasing decisions, or by third-party organizations to audit reviewing websites. Finally, we confirm the applicability of the veracity effect and the chance level of human detection accuracy to the online review context (Bond & DePaulo, 2006; Levine et al., 1999; Vrij, 2000). Moreover, we illustrate the presence of fake online reviews on reviewing websites and the possibility of detecting them using an automated algorithm.

In terms of detection, we confirm that linguistic deception is a useful method for identifying deceit (Zhou et al., 2004) and that, for better accuracy, cues should be taken as clusters and not individually (e.g., Fuller et al., 2011). Furthermore, we show that Coh-Metrix is a good tool for linguistic analysis (Graesser et al., 2011) and could be an alternative to the widely used LIWC, as it analyzes language on a deeper level. It should be noted that automated deception detection is an important direction for academic research and a practical solution for both consumers who wish to verify information and review websites that with to control the quality of the information on their sites.

Our results, like those of previous studies, clearly show that individuals are inaccurate at detecting the veracity of online reviews (Bond & DePaulo, 2006; Levine et al., 1999; Vrij, 2000). Furthermore, the truth bias exists, and it is significant (Levine, 2014; Levine et al., 1999). This demonstrates consumers' inability to detect fake online reviews and the vulnerability this creates. Although consumers evaluate the grammar and the written style of an online review to judge its credibility (Ketron, 2017), micro-linguistic deception cues are difficult for humans to identify and apply. This leads us to conclude that individuals need some external support, such as an automated detection tool, to evaluate the nature of online reviews.

The results of our investigation into the quality of online reviews on the three most-used restaurant reviewing websites might somewhat contradict commonly held rules for detecting fake online reviews (e.g., http://www.wikihow.com/Spot-a-Fake-Review-on-Amazon); however, they are in line with the results of most previous studies. For instance, we might suggest that a "good" active reviewer profile is not a guarantee of a genuine review because some sophisticated spammers can create and sustain seemingly trustworthy profiles (Wang, Xie, Liu, & Yu, 2011). In addition, the length of a review depends a great deal on the functionalities of the website and community management. When long reviews are rewarded, individuals might tend to write long reviews; however, when only limited feedback is requested, individuals might write only short passages. Suspicion should be raised when the structure of a genuine-looking review, including the profile and the picture, is very similar to those of other posts on the same product.

Many studies have suggested detecting deception by focusing on semantic and non-semantic similarities (e.g., Lau, Liao, & Xu, 2010). Furthermore, as suggested by prior studies, individuals often decide to publish true reviews after either very good or very bad experiences (e.g., Hennig-Thurau et al., 2004); therefore, and in contrast to the conclusions of Luca and Zervas (2016), extreme accounts are not always fake. Moreover, spammers know that very good and very bad reviews are often eliminated. They therefore "smooth out" their reviews with moderate ratings (i.e., two or four stars out of five). Interestingly, we found that most of the suspicious reviews had been written for the same restaurants. This might support ideas related to detecting deception-prone offerings (Li et al., 2013) and tracing economic incentives (Luca & Zervas, 2016). Moreover, we confirmed that single-post profiles (Xie, Wang, Lin, & Yu, 2012), reiterating posts (on different websites or in different languages), and overly general feedback were potentially signs of spam. Finally, the percentage of suspicious reviews in our study, 11.4% to 14.14%, was close to that found by Hu, Bose, Koh, and Liu (2012).

6.2. Managerial implications

Two of our main findings—consumers' incapacity to detect fake online reviews on their own and the apparent presence of such reviews online—lead to the logical conclusion that online platforms that host reviews should pay greater attention to content quality. Filtering submitted reviews with automated tools might be one solution.

Although we confirmed that people are naturally trustful and have a truth bias, they also give less credibility to online communications (e.g., Zhang et al., 2016). Our findings show that a quality label might recomfort consumers and reestablish the truth bias, which is encouraging for reviewing platforms. Thus, additional information, such as "trusted reviews" or "verified purchase", might improve the trustworthiness of website content.

Automated filters pointing out suspicious content could also be used by consumers, and simple algorithms could be integrated directly into Internet browsers and mobile applications. Finally, legislators and other third parties could use the tool develop here to conduct external audits of reviewing websites.

Companies can fall victim to fake online reviews in three different ways: They might be targets of negative reviews, their competitors might benefit from an astro-turfed online reputation and undermine their visibility and attractivity (Lappas, Sabnis, & Valkanas, 2016), or consumers might become over-suspicious and doubt authentic positive feedback. To avoid the risks related to fake reviews, companies should: monitor their online reputation, collaborate with online platforms that verify posted content (Mayzlin et al., 2014), and be aware of legal procedures and possible actions against deceivers (e.g., Hunt, 2015).

6.3. Limitations and directions for further research

We are aware of some limitations related to the experimental design of this research. The respondents had to imagine themselves in a described situation, and although the realism of the scenario was found satisfactory, analyzing consumers in real-life situations would be more insightful. Moreover, further studies could investigate retrieving linguistic cues from other contexts (e.g., different products and services). It would also be interesting to test the robustness of the algorithm across different communication media (e.g., emails, online chats, voicemail). This would facilitate an evaluation of the universality of the approach and the cues.

There is also a promising field of deception detection research regarding human abilities. First, it would be interesting to understand the types of information people tend to misperceive. We rated the detection success for every review, but the small number (10) used for the general evaluation of detection capabilities was not sufficient to compare the various characteristics of the messages (e.g., positive vs. negative, short vs. long, or emotional vs. factual). Previous studies have shown that consumers evaluate not only textual (e.g., Ketron, 2017), but also contextual indicators to determine the trustworthiness of online reviews (Munzel, 2016). Therefore, future studies should account for the effects of context (e.g., the platform and product (Park & Lee, 2009)) on humans' capacity to detect deceptive content.

Although we are motivated by the ease of use and the performance of our detection tool, we believe that it can be refined. First, we might use the same data and indices, but apply a different decision method. Neural networks, for example, have been shown to yield better results than discriminant analysis, logistic regression, and decision trees (Zhou et al., 2004). We believe in the applicability of text-based linguistic analysis (Mukherjee et al., 2013), which could be even further improved with the latest advances in linguistic software (Feng, Banerjee, & Choi, 2012). Thus, analyzing language on a "nano" level might allow for more accuracy and less context bias. However, detection does not have to be text-based (Wang et al., 2011). There are new approaches based on ratings (Feng, Xing, et al., 2012; Lim, Nguyen, Jindal, Liu, & Lauw, 2010); networks involving brand, product, and reviewer (Akoglu, Chandy, & Faloutsos, 2013); and offerings themselves (Li et al., 2013). Furthermore, we should consider different types of deceivers (e.g., single malefactors vs. organized groups; professionals vs. novices). Previous studies have shown that using a combination of different techniques and methods improves accuracy (Fuller et al., 2011). In addition, we saw that reviews and reviewing communities vary based on the functionalities of a website. Therefore, detection approaches will likely need to be adjusted for each context. Furthermore, computer-generated fake reviews continue to become increasingly sophisticated and difficult to detect (Yao, Viswanath, Cryan, Zheng, & Zhao, 2017). Hence, algorithms should be adjusted to detect deceptions created by artificial intelligence. Finally, no detection tool can guarantee 100% accuracy.

Declarations of interest

None.

Appendix A. Linguistic differences among truthful (T) and deceptive (D) online reviews

LIWC cues	Positive reviews	Negative reviews	Coh-Matrix cues	Positive reviews	Negative reviews
Summary language variables			**Descriptive**		
Analytical thinking	T > D*	T < D*	Word length, number of syllables (mean)	T < D*	–
Authentic	T < D**	T > D***	Word length, number of syllables (number of letters)	T < D*	–
Emotional tone	T > D***	T < D***	Number of paragraphs	*T > D**	*T > D**
Words > 6 letters	T > D*	T < D***	Lengths of paragraphs (sentences count)	T > D*	–
Dictionary words	–	T < D***	**Text Easability** (cohesion)	*T < D**	*T < D**
Linguistic dimensions			Temporality	–	*T > D***
Total function words (e.g., it, to, no, very)	T < D***	–	**Referential cohesion**		
Personal pronouns	–	T < D*	Noun overlap adjacent sentences (mean)	–	T < D*
1st personal plural (we)	T < D**	–	Stem overlap adjacent sentences (mean)	–	T < D**
2nd person (you)	–	T < D*	**All connectives**	T > D*	–
3rd person singular (she/he)	T > D*	–	Logical connectives	*T > D**	*T > D***
Prepositions	–	T < D*	Adversative and contrastive	*T > D***	*T > D***
Auxiliary verbs	T < D**	–	Temporal connectives	*T > D***	*T > D***
Adverbs	T < D*	T > D**	1st personal plural (we)	T < D*	–
Conjunctions	T > D**	–	2nd person pronoun	T < D*	–
Common verbs	T < D**	–	**Syntax and situation model**		
Negations	T < D***	T > D**			
Numbers	T > D**	–	Sentence syntax similarity	–	T < D*
Psychological processes			Agentless passive voice	–	T < D*
Positive emotion	*T > D****	*T < D****	Content words	–	T < D*
Negative emotion	*T < D****	*T > D****	Imagibility for content words	T < D**	–
Anxiety	T < D*	T > D*	Hypernymy for nouns (mean)	T < D*	–
Friends	T > D*	T < D*	Hypernymy for verbs (mean)	T < D**	T < D*
Differentiation (but, else)	–	T > D***	Verbs	–	T > D*
Time	T < D**	T > D**	Adjectives	*T < D***	*T < D***
Leisure	–	T < D*	Gerunds	T < D*	–
Netspeak (bthw, lol, thx)	T > D*	–			
All punctuation	–	T > D*			

* p-Value < .1.
** p-Value < .01.
*** p-Value < .001.

Appendix B. ROC and Brier score computed for the retained detection algorithm

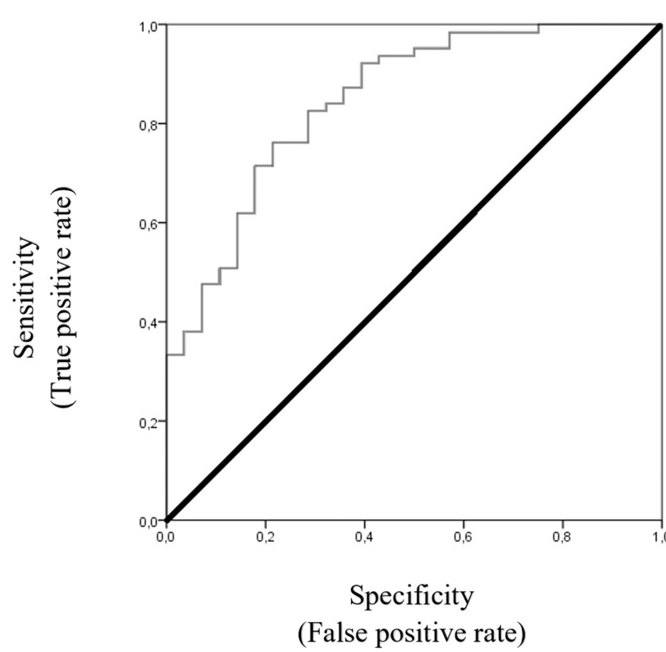

Fig. B.1. ROC for the retained detection algorithm.

Appendix C. Random Forest output for fake online reviews detection

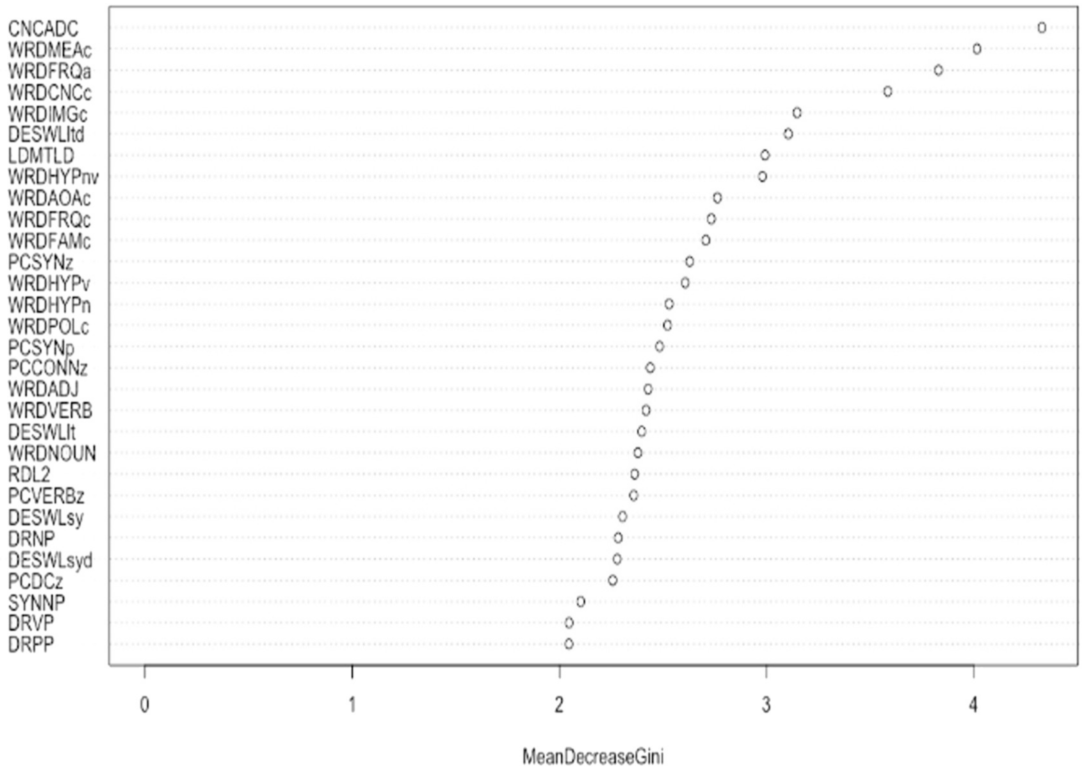

Fig. C.1. Random Forest output for positive online reviews.

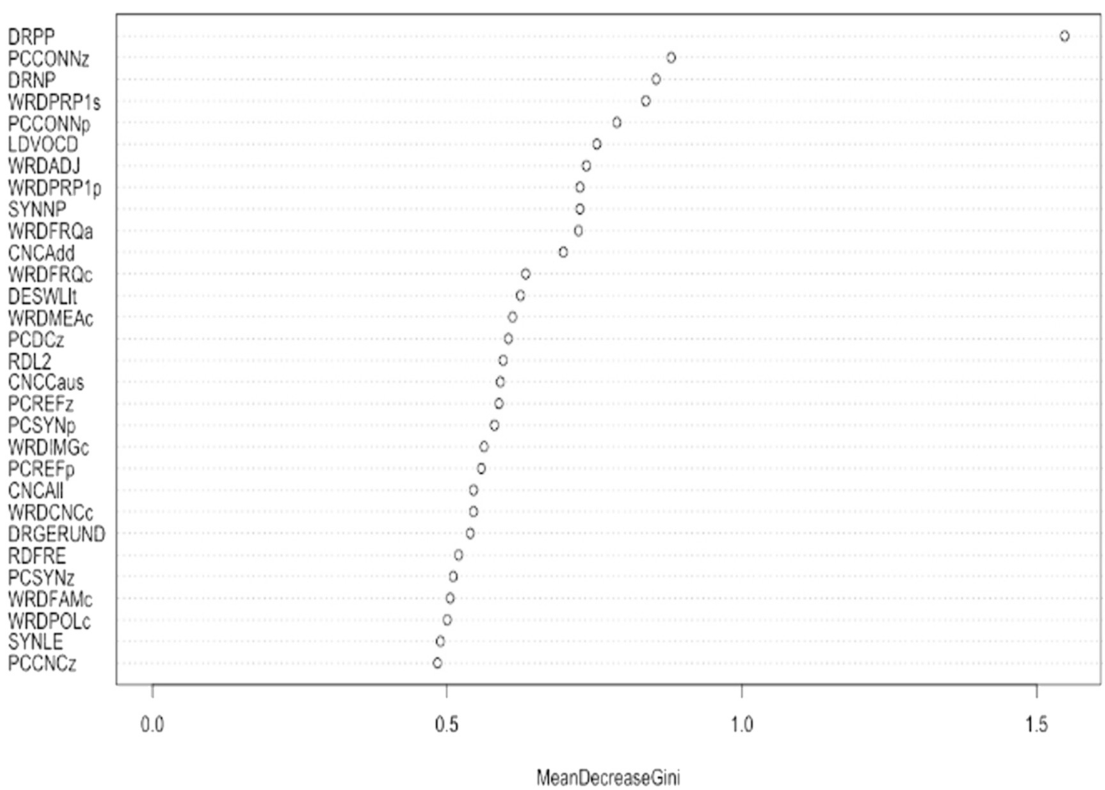

Fig. C.2. Random Forest output for negative online reviews.

Appendix D. An excerpt of the set of linguistic cues for detecting fake reviews

Coefficient (in a binominal regression, where 0 is true and 1 is a fake review)	Explanation	Code
Descriptive linguistic cues		
−0.184**	The average number of sentences per paragraph within the text.	DESPL
−0.582**	The total number of paragraphs in the text.	DESPC
Conclusion: Deceiving messages have fewer paragraphs and sentences per paragraph.		
Text easability		
312.282*	Referential cohesion: overlapping words and ideas across sentences and the entire text, forming explicit logical chains, uniting the text and making it more sensible.	PCREF
128.953*	Concrete words: words are more precise and meaningful and evoke associations and mental images more easily. This makes the text easier to understand and process.	PCCNC
−0.019*	Time preciseness.	PCTEMP
Conclusion: overall, deceptive messages are very structured, coherent, and easy to understand. However, they slightly lack temporal precision.		
Syntactic pattern density		
0.589*	The incidence of adjectives.	WRDADJ
−0.502*	The incidence score of temporal connectives.	CNCTemp
−0.494*	The incidence of intentional verbs (actions and events)	SMINTEp

* p-Value < .1.
** p-Value < .01.

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Daria Plotkina, Associate Professor at EM Strasbourg Business School, University of Strasbourg, obtained her doctorate in Digital Marketing from EM Strasbourg Business School. Her research focuses on online information and interaction, was communicated on international conferences and is published in Journal of Retailing and Consumer Services.

Andreas Munzel, Associate Professor at Toulouse School of Management (TSM). He obtained a doctorate in Management Sciences from the University of Munich (LMU) in Germany. His work focuses on online social interactions and their implications for marketing and is published in international scientific journals, such as Journal of Service Management, Journal of Retailing and Consumer Services, International Journal of Electronic Commerce, and Technological Forecasting and Social Change.

Jessie Pallud, Full Professor at EM Strasbourg Business School, University of Strasbourg. She obtained her PhD in Management of Information Systems from Paris Dauphine University and Georgia State University. Her research focuses on the adoption of information systems in the workplace and human-computer interaction. She has also conducted research on user experience in cultural heritage settings, namely museums. Jessie Pallud is published in Information and Management, Journal of Management Information Systems, and European Journal of Information Systems.